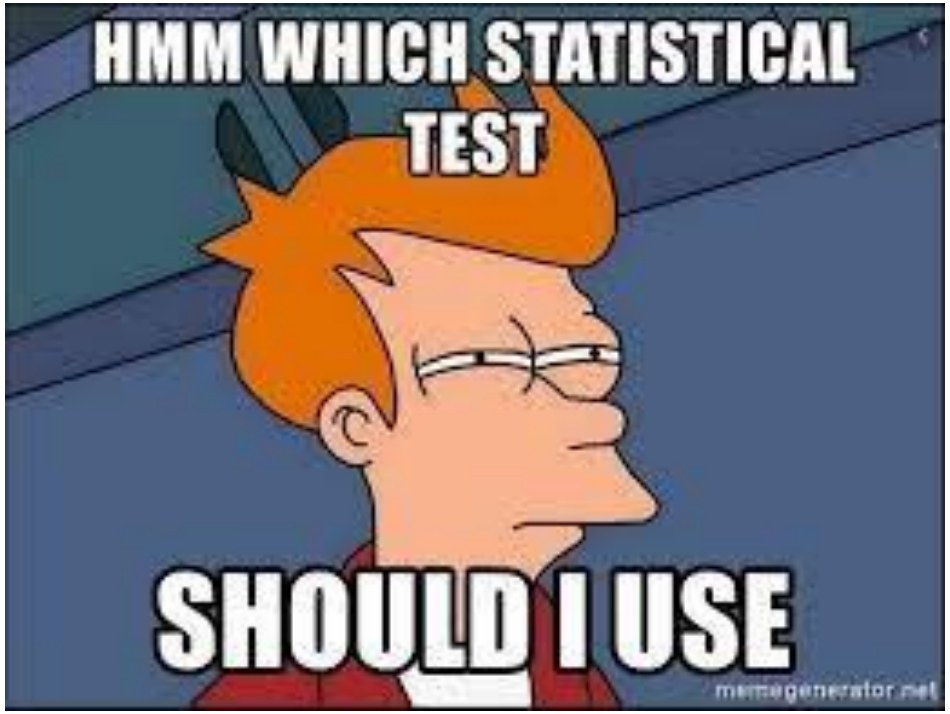
Chapter 10

Two Sample Tests of Hypotheses

Part 2 – Dependent Samples

(Text ref 10.5 - 10.6)



Key Concepts (this part)

- Use and understand the formulas in this chapter
- Know the difference between independent and dependent samples
- Construct hypothesis tests for dependent samples
  - Paired t-test
  - Paired Wilcoxon Signed Rank Test

Remember Terminology

- Two samples are *independent* if they come from different, unrelated populations.
  - E.g. Mean wage of plumbers vs. Mean wage of electricians

Focus for these slides

- Two samples are *dependent* if they come from matching or paired observations or if they come from 2 measurements of the same objects, separated by intervention.
  - E.g. Weight loss program before and after measurements

Recap

Two Dependent samples *t* Test (Paired t)

A	B	d
A_1	B_1	$A_1 - B_1$
		$A_2 - B_2$
A_3	A_3	$A_3 - B_3$

Single sample #s

Difference of means:

CH 10 Pg 1 → $\mu_1 - \mu_2$

→ This is how we "pair"

- Look for samples that have related measurements
  - E.g. before and after weights of people
  - E.g. Student grades on Test and Final exam
- Standard deviation of populations always unknown
- We are interested in mean difference of the samples/populations
- The test statistic is

$$t = \frac{\bar{d} - \mu_d}{\frac{s_d}{\sqrt{n}}}, \qquad df = n-1$$

μ_d

Sample

CH 9: $t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$

Where $\bar{d}$ is the mean of the differences between the samples, s_d is the standard deviation of the differences and n is the number of pairs (differences)

Dependent/Paired tests Example

RAW DATA * STEPS

Five individuals went on a special diet for 3 weeks. They weighed themselves before starting the diet and after completing it. Their weights (in kg) were recorded. Is there sufficient evidence at the 2.5% level to suggest that the average weight loss of this group was more than 1 kg?

$\alpha = 0.025$

Person	Before	After
Arnold	102	97
Jason	77	77
Sylvester	92	93
Dwayne	108	100
Mark	87	78

→ 2 SAMPLES

→ DATA DEPEND ON PERSON

Step #1 State null and alternate hypothesis

LET μ_d REPRESENT THE MEAN DIFFERENCE IN WEIGHTS FOR PEOPLE ON THIS SPECIAL DIET (d = BEFORE - AFTER)

$$H_0: \mu_d \leq 1$$

$$H_a: \mu_d > 1$$

Step #2: Select the level of significance
Step # 3 – Identify the Test statistic

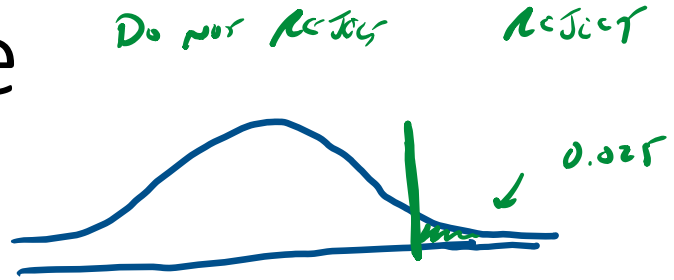
$\alpha = 0.025$

$$t = \frac{\bar{d} - \mu_d}{s_d / \sqrt{n}}$$

USE DEPENDENT t

WHY? 2 DEPENDENT SAMPLES

Step #4: State the Decision Rule



H_a : "$>$" → right sided

$\alpha = 0.025$

USE t → TABLE IV

↳ $df = n-1$

$= 5-1 = 4$

so $t^* = 2.776$

Decision Rule: Reject H_0 if $t_{calc} > 2.776$

Step #5 – Make a Decision

$\bar{d} = 4.2$

$\mu_d = 1$ (STEP 1)

$s_d = 4.5497$

$n = 5$

$$t_{CALC} = \frac{\bar{d} - \mu_d}{s_d / \sqrt{n}}$$

$$t_{CALC} = \frac{4.2 - 1}{\frac{4.5497}{\sqrt{5}}}$$

$$t_{CALC} = 1.5727$$

Person	Before		After		DIFF (d)
Arnold	102	−	97	=	5
Jason	77	−	77	=	0
Sylvester	92	−	93	=	−1
Dwayne	108	−	100	=	8
Mark	87	−	78	=	9

SINCE $t_{CALC} = 1.57 < 2.776$ → DO NOT REJECT H_0

THERE IS INSUFFICIENT EVIDENCE TO SUGGEST THE MEAN WEIGHT LOSS FOR THIS SPECIAL DIET IS MORE THAN 1 KG.

Paired t Test Meets Technology

In a report on “Peak discharge from Unlogged and Logged Watersheds”, the researchers compiled (paired) data on peak discharge from storms in two watersheds, one unlogged and one logged (100% clear-cut). If there is an effect due to clear cutting, one would expect that the runoff would be greater in the logged/clear-cut area than in the unlogged area. The runoffs, in cubic meters per second per square kilometer are provided in the Excel file *ClearCut.xls*.

Assume that the distribution of the differences are normally distributed and apply the Paired t-test with a 0.05 level of significance to perform the required hypothesis.

α

Paired t Test Meets Technology

Sketch of solution:

STEP 1 LET d = LOGGED − UNLOGGED

$H_0: \mu_d = 0$

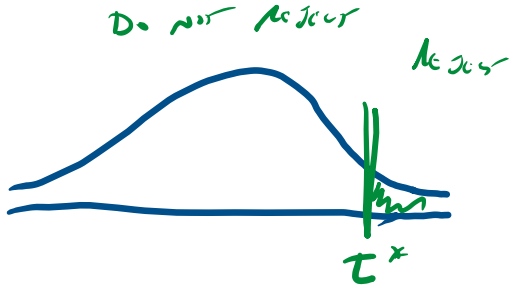
$H_a: \mu_d > 0$

STEP 2 USE $\alpha = 0.05$

STEP 3: USE $t = \dfrac{\bar{d} - \mu_d}{s_d/\sqrt{n}}$

2 SAMPLES, PAIRED AREAS

STEP 4:



$df = 352 - 1 = 351$

[NOT ON TABLE IV → BE CONSERVATIVE AND USE $df = 300$]

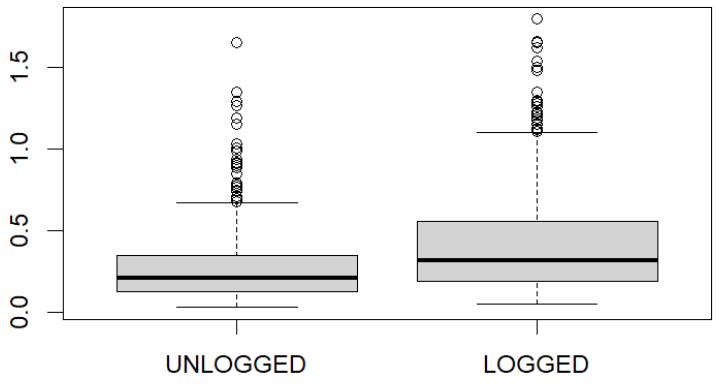
$t^* = 1.650$

DECISION RULE: REJECT H_0 IF

$t_{CALC} > 1.650$

Paired t Test Meets Technology

Sketch of solution:




```
> loggeddiff = ClearCut$LOGGED - ClearCut$UNLOGGED
> t.test(loggeddiff, alternative = "greater",mu = 0, conf.level = 0.95)

        One Sample t-test

data:  loggeddiff
t = 20.501, df = 351, p-value < 2.2e-16
alternative hypothesis: true mean is greater than 0
95 percent confidence interval:
 0.1379331       Inf
sample estimates:
mean of x
     0.15
```

$t_{calc} = 20.501 > 1.65 \rightarrow$ REJECT H_0

ALSO $p = 2.2 \times 10^{-16} < \alpha = 0.05 \rightarrow$ REJECT H_0

THERE IS SUFFICIENT EVIDENCE TO SUGGEST THAT THE RUN-OFF IS GREATER IN LOGGED/CLEAR CUT AREAS.

CH 9: WILCOXON SIGNED RANK TEST (WSRT)

CH 10 pt 1: MANN-WHITNEY-WILCOXON

→ NON PARAMETRIC

Two Dependent samples Paired Wilcoxon Test

- Look for samples that have related measurements
  - E.g. before and after weights of people
  - E.g. Student grades on Test and Final exam
- Samples come from populations with symmetric differences WSRT (CH 9)
- We are interested in mean difference of the samples/populations μ_d
- The test statistic is non-parametric

$$W_A = sum\ of\ the\ positive\ ranks$$

CH 9 → TABLE V

→ RANKED SAMPLE DIFFERENCES

→ W_{CALC}

Two Dependent samples Paired Wilcoxon Test

(CH 9)

- The 'quirks' about the Wilcoxon Signed Rank Test still hold:
  - Any sample values equal to the hypothesized mean difference (μ_d) should be removed from the sample, and recompute
    n' = n – number sample values equal to μ_d (HYPOTHESIZED VALUE)
  - Any ties should be assigned the mean of the ranks they would have been assigned, assuming there were no ties.
  - For a left tailed (or for the left tail of a 2-tailed test), use Table V to find W_α with sample size n', then compute

$$W_{1-A} = \frac{n'(n'+1)}{2} - W_\alpha$$

DEPENDENCY ON CITY

Dependent/Paired Example

RAW DATA

An independent testing agency is comparing the daily rental cost for renting a compact car from Hertz and Avis. A random sample of 8 Canadian cities revealed the following information. At the 0.05 significance level can the testing agency conclude that there is a difference in the rental amount charged?

Assume the sample differences come from a population with a symmetric distribution.

City	Hertz ($)	Avis ($)
Saint John	42	40
Halifax	45	43
Montreal	48	48
Toronto	56	52
Edmonton	37	32
Calgary	45	48
Vancouver	41	39
Victoria	46	50

Step #1 State null and alternate hypothesis

LET d = HERTZ \$ - AVIS \$

$$H_0: \mu_d = 0$$

$$H_a: \mu_d \neq 0$$ ("DIFFERENCE" IN RENTAL PRICES)

Step #2: Select the level of significance
Step # 3 – Identify the Test statistic

$\alpha = 0.05$

USE PAIRED WILCOXON TEST

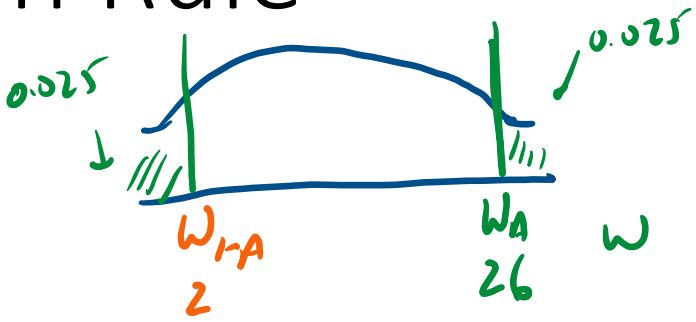
→ 2 SAMPLES THAT "DEPEND" ON CITY

Step #4: State the Decision Rule

H_a : "$\neq$" → 2 SIDED TEST

$\alpha = 0.05$ → $\alpha/2 = 0.025$

USE W → TABLE V



$W_A[0.025, 7] = 26$ (TABLE V)

$n = 8$ (CISIKS)

$n' = 7$ (ONE VALUE = 0)

LEFT TAIL W_{1-A} : $\frac{n'(n'+1)}{2} - 26$ (W_A)

$= \frac{7(8)}{2} - 26 = 2$

<u>DECISION RULE</u>: REJECT H_0 IF $W_{CALC} < 2$ OR IF $W_{CALC} > 26$

Step #5 – Make a Decision

SMALLEST TO LARGEST

A IS BASED ON

City	Hertz ($)		Avis ($)		diff	|diff|	Rank	Signed Rank
Saint John	42	-	40	=	2	2 *	2	2
Halifax	45	-	43	=	2	2 *	2	2
Montreal	48	-	48		0			
Toronto	56	-	52		4	4 **	5.5	5.5
Edmonton	37	-	32		5	5	7	7
Calgary	45	-	48		-3	3	4	-4
Vancouver	41	-	39		2	2 *	2	2
Victoria	46	-	50		-4	4 **	5.5	-5.5

* RANK 1, 2, 3 TIED IN VALUE:

$$\frac{1+2+3}{3} = 2$$

** RANK 5, 6 TIED IN VALUE:

$$\frac{5+6}{2} = 5.5$$

W_{CALC} = POSITIVE RANKS: $2 + 2 + 5.5 + 7 + 2$

$= 18.5$

SINCE $W_{CALC} = 18.5$ BETWEEN 2 & 26, WE DO NOT REJECT H_0.

Step #5 (continued)

CONCLUSION: THERE IS INSUFFICIENT EVIDENCE TO SUGGEST THAT THERE IS A SIGNIFICANT DIFFERENCE IN RENTAL CAR COSTS BETWEEN HERTZ & AVIS FOR THIS TYPE OF VEHICLE.

Paired Wilcoxon Meets Technology

In a report on “Peak discharge from Unlogged and Logged Watersheds”, the researchers compiled (paired) data on peak discharge from storms in two watersheds, one unlogged and one logged (100% clear-cut). If there is an effect due to clear cutting, one would expect that the runoff would be greater in the logged/clear-cut area than in the unlogged area. The runoffs, in cubic meters per second per square kilometer are provided in the Excel file *ClearCut.xls*.

Assume that the distribution of the differences come from a symmetric distribution and apply the Paired Wilcoxon test at the 0.05 level of significance to perform the required hypothesis.

Paired Wilcoxon Meets Technology

Sketch of solution: (SAME AS SLIDE 11)

STEP 1: LET d = LOGGED − UNLOGGED

$H_0: \mu_d = 0$

$H_a: \mu_d > 0$

STEP 2: $\alpha = 0.05$

STEP 3: USE PAIRED WSRT

STEP 4: DECISION RULE: REJECT H_0 IF $p < \alpha$

STEP 5:

```
> wilcox.test(loggeddiff, alternative = "greater",mu = 0, conf.level = 0.95)

        Wilcoxon signed rank test with continuity correction

data:  loggeddiff
V = 61297, p-value < 2.2e-16
alternative hypothesis: true location is greater than 0
```

WCALC

P-VALUE

SINCE P-VALUE $= 2.2 \times 10^{-16} < 0.05$, WE REJECT H_0.